

Influence of Tiered Starting Positioning: Formula 1 British Grand Prix (Silverstone)



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CMSE 201

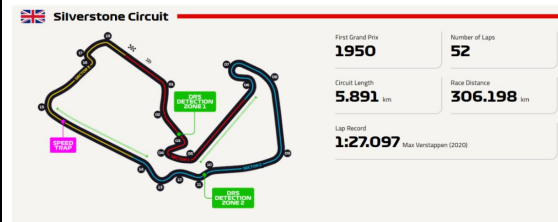
Workflow

1. Import, clean and merge data
2. Calculations
3. Visualization Ideas
4. Conclusion

Assumptions

- Historical trends are reliable indicators.
- Crashes, mechanical failures, and weather unpredictability are not modeled.
- Data assumes team stability per race (no mid-season driver switches unless tracked).

Formula 1 British Grand Prix(Silverstone) Podium Prediction(Team based)



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#load in 2019-2024 drivers and result
drivers2019 = pd.read_csv('formula1_2019season_drivers.csv')
driver2020 = pd.read_csv('formula1_2020season_drivers.csv')
driver2021 = pd.read_csv('formula1_2021season_drivers.csv')
driver2022 = pd.read_csv('formula1_2022season_drivers.csv')
driver2023 = pd.read_csv('formula1_2023season_drivers.csv')
driver2024 = pd.read_csv('formula1_2024season_drivers.csv')

race_results2019 = pd.read_csv('formula1_2019season_raceResults.csv')
race_results2020 = pd.read_csv('formula1_2020season_raceResults.csv')
race_results2021 = pd.read_csv('formula1_2021season_raceResults.csv')
race_results2022 = pd.read_csv('formula1_2022season_raceResults.csv')
race_results2023 = pd.read_csv('formula1_2023season_raceResults.csv')
race_results2024 = pd.read_csv('formula1_2024season_raceResults.csv')

#print(driver2024.head())
#print(race_results2024.head())
```

1.

```
#extract only track = Great Britain

#Clean missing values
# Drop any fully empty rows
drivers2019.dropna(how='all', inplace=True)
race_results2019.dropna(how='all', inplace=True)
```

2. Final Data Format

Track	Team	Avg Start	Avg Finish	Fastest Laps	Rolling Points
AUS	Ferrari	4.0	4.5	0	25

Research Questions:

Does a driver's starting position impact race results?

Additional questions:

- Which team improved their position the most?
- Which team accumulated points most steadily?
- Was the driver who set the fastest lap one of the drivers (team) on the podium?

Features:

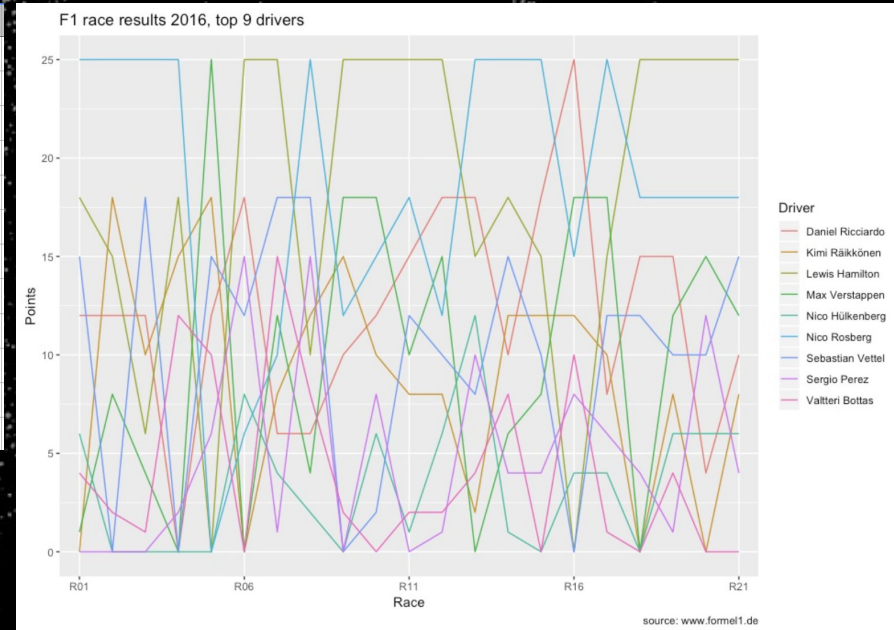
Avg Start, Avg Finish, Fastest Laps, and Rolling Points

Visualizations

- **Compare Average Start and Finish Positions (Per Team)**
- **Points Progression Across Races**
- **Scatter plot:** Start position vs. Finish position, change in position.

Point - Position relationship

Position	Points
1	26
2	18
3	15
4	12
5	10
6	8
7	6
8	4
9	2
10	1
11	0
12	0



3. Example points progression plot